

$$z = \frac{1}{2}at^2$$

$$\frac{4}{a} = t^2$$

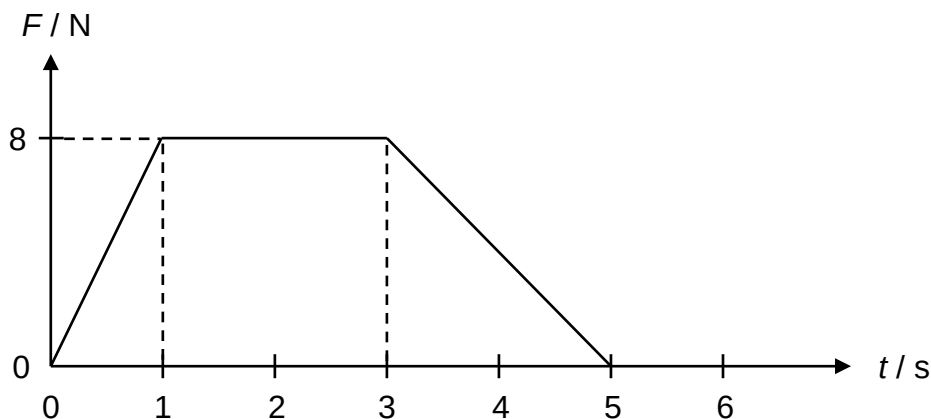
$$t = \frac{2}{\sqrt{a}} \text{ ----- (1) where } t \text{ is the time taken for stone to fall on Earth}$$

Acceleration of free fall of moon = $\frac{a}{6}$, where a is the acceleration of free fall on Earth.

Sub $\frac{a}{6}$ into equation (1)

$$t_{\text{moon}} = \frac{2}{\sqrt{\frac{a}{6}}} = \frac{2}{\sqrt{a}}(\sqrt{6}) = t\sqrt{6}$$

- 4 An object of 2 kg initially at rest, is acted on by a force F . The variation with time t of F is as shown below.



What is the momentum of the object at time $t = 6$ s?

A	0 Ns	B	14 Ns	C	28 Ns	D	56 Ns
----------	------	----------	-------	----------	-------	----------	-------

Answer: C

The area under the F - t graph represents the change in momentum, Δp experienced by the object.

$$\Delta p = \frac{1}{2} \times (5 + 2) \times 8$$

$$\Delta p = 28$$

$$p_f - p_i = 28$$

Since object starts from rest,

$$p_f - 0 = 28$$

$$p_f = 28 \text{ N s}$$